

Figure 8(a): High-Confidence Empirical Test Classifications (MobileNetV2 >99% Posterior Confidence)

Representative holdout test samples correctly classified with peak certainty on 4,500 holdout images.

CORRECT [TN] | Pred: Cat (99.8%)



True: Cat

CORRECT [TN] | Pred: Cat (99.5%)



True: Cat

CORRECT [TP] | Pred: Dog (99.8%)



True: Dog

CORRECT [TP] | Pred: Dog (99.6%)



True: Dog

Sharp triangular ear geometry and prominent whisker patterns detected. Feline facial contour and horizontal eye alignment correctly classified. Distinctive canine nasal bridge and elongated muzzle structure recognized. Canine cranial morphology and characteristic ear posture identified.

Figure 8(b): Error Topology & Decision Boundary Failure Analysis (Diagnostic Edge Cases)

In-depth diagnostic breakdown of failure modes: low illumination, unusual sleeping postures, and partial occlusions.

FALSE POSITIVE [FP]



True: Cat | Pred: Dog (62.4%)

Long-haired Persian cat in low lighting; fluffy fur obscuring ear tips mimics small canine facial geometry.

FALSE NEGATIVE [FN]



True: Dog | Pred: Cat (61.8%)

Small curled puppy resting on cushion; elliptical body posture and hidden snout resemble sleeping cat.

HARD TRUE POSITIVE



True: Dog | Pred: Dog (78.5%)

Backlit dog against high-glare background; network successfully isolates invariant anatomical silhouette.

HARD TRUE NEGATIVE



True: Cat | Pred: Cat (86.2%)

Cat with partial foreground occlusion; deep receptive fields correctly resolve visible eye and ear geometry.